

Original article

A hybrid artificial intelligence and deep learning architecture for accurate renewable energy forecasting: comprehensive case studies on wind and PV power

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ABSTRACT

This study proposes a novel hybrid forecasting architecture that synergistically integrates artificial intelligence and deep learning techniques. This approach effectively overcomes the limitations of conventional single models in characterizing complex weather-power coupling relationships. First, we propose an enhanced honey badger algorithm based on self-learning factors, which incorporates three key innovations: (1) the good point set principle for population initialization, (2) an adaptive dynamic learning mechanism, and (3) a Gaussian mutation strategy. Comprehensive case studies demonstrate that the proposed algorithm achieves superior performance in iterative optimization compared to existing methods. Second, we develop a short-term forecasting model for wind and photovoltaic power generation, explicitly designed to balance model effectiveness and generalizability. The self-learning honey badger algorithm is employed to optimize model parameters, enabling robust feature extraction from renewable energy output sequences. Finally, extensive validation and analysis are conducted using real-world hybrid power generation datasets combining wind and solar energy. The results show that the proposed model achieves the lowest root mean square error values and consistently exceeds 99% in R2 across diverse weather conditions. These findings provide a scientific foundation for enhancing the economic operation and dispatch of renewable energy generation systems.

Introduction

As the global energy transition continues to deepen, China's installed power generation capacity reached 3.49 billion kW by April 2025, with wind and photovoltaic (PV) power accounting for more than one-third of the total. Specifically, PV and wind capacities reached 990 million kW and 540 million kW, increasing year-on-year by 47.7 % and 18.2 %, respectively [1,2]. However, the rapid growth of wind and PV penetration has intensified challenges to power system stability. Owing to strong dependence on meteorological conditions, wind and PV generation exhibit significant intermittency and volatility [3], which significantly increases peak shaving demand, exacerbates renewable energy curtailment, and leads to substantial energy waste [4]. Meanwhile, power system operation faces growing pressure from real-time source-load balancing and escalating regulation costs [5].

High precision wind and PV power forecasting is the core means for mitigating the randomness and volatility of renewable generation and is strategically important for building new power systems and ensuring energy security [6]. Accurate short-term forecasting enhances generation predictability, supports refined dispatch planning, and improves the controllability of renewable resources through real-time regulation systems such as automatic generation control [7]. Moreover, advanced forecasting technology is essential for alleviating renewable energy integration challenges and serves as a cornerstone for building a renewable-dominated power system, thereby supporting the national goals of carbon peaking and carbon neutrality.

Literature review

Renewable energy generation forecasting is categorized into long-

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term (several months), medium-term (several weeks), short-term (a few days) and ultra-short-term (from several minutes to a few hours) forecasting based on time scales [8,9]. Among these, short-term forecasting plays a pivotal role in power system dispatch and operation, as its accuracy directly influences scheduling decisions and economic performance [10]. High precision short-term forecasting not only optimizes generation planning and reduces reserve requirements, but also enhances the grid's capability to integrate intermittent renewable energy, making accuracy and robustness key research challenges [11].

Existing short-term forecasting methods are generally categorized into physical model-based and statistical model-based approaches. Hu et al. [12] constructed a short-term prediction method for solar radiation by introducing a clear sky radiation model based on atmospheric physics principles, providing a theoretical basis for flexible regulation of photovoltaic power generation system. Yeom et al. [13] estimated the availability of solar irradiance on the land surface of North Korea using a physical model based on COMS satellite data and atmospheric composition statistical data. Meanwhile, compared with the measured data from the meteorological tower, numerical weather forecasting (NWP) was used to analyze the wind speed data of the wind farm. This study validated the reliability of satellite and numerical model data. Zhao et al. [14] proposed a new method for correcting wind speed prediction errors by combining mathematical analysis with physical models, which significantly improved prediction accuracy while maintaining the advantages of the physical model. While these approaches provide strong physical interpretability, their performance depends heavily on large scale, high quality meteorological data, resulting in high computational cost and sensitivity to weather uncertainty, which limits their practical applicability in short-term wind and PV forecasting [15].

Statistical model-based methods predict future generation by mining historical data patterns and probability distributions, offering advantages such as simpler structure, wide applicability and stronger generalization [16]. These methods include time series analysis [17], machine learning [18], and deep learning approaches [19]. With advances in data availability and computing power, machine learning techniques such as support vector machine (SVM), extreme learning machine (ELM), gaussian process (GP), have demonstrated strong nonlinear modeling capability in renewable energy forecasting [20–22].

Compared with traditional machine learning, deep learning models leverage hierarchical structures for automatic feature extraction and nonlinear transformation, significantly improving representation capability and generalization in complex, high dimensional scenarios [23,24]. Consequently, recurrent neural networks such as long short term memory (LSTM), gated recurrent unit (GRU) and their bidirectional variants have become benchmark models for time series forecasting [25–27]. In response to the problem of insufficient prediction accuracy caused by NWP wind speed errors in short-term wind power forecasting, Wang et al. [28] proposed a short-term prediction model that integrates kepler optimization algorithm and convolutional neural network long short term memory network. Sarma et al. [29] proposed a meta learning method combined with multiple deep learning models to improve the prediction accuracy of PV power generation. Experiments show that this method is better than a single model on multiple PV systems. Zhao et al. [30] proposed a high-precision wind power forecasting method by combining wind power output data, feature extraction techniques, and a GRU model, which resulted in a MAPE value of 11.62 %.

Although deep learning models have significant advantages in wind and PV power forecasting, their training and testing results are highly sensitive to random parameters, which directly affects the accuracy and stability of the model [31,32].

Research gaps and proposed solutions

Existing studies reveal several unresolved challenges in deep learning-based renewable energy forecasting. Firstly, the non-stationary

and stochastic nature of renewable generation, coupled with the complex, high-order nonlinear relationships between meteorological variables and power output, poses significant difficulties for accurate time-series modeling. While deep learning models possess the capacity to capture these patterns, their performance is highly sensitive to hyperparameters and initial random weights, leading to instability and sub-optimal results in practical applications. Moreover, adapting models to site-specific wind or PV plants usually requires extensive manual tuning, resulting in a low efficiency “one-off” optimization paradigm that limits large-scale deployment. The absence of an automatic and adaptive optimization mechanism therefore remains a key research gap.

To address these challenges, this study proposes an adaptive short-term forecasting framework that integrates an intelligent optimization algorithm with deep learning. A self-learning factors-based honey badger algorithm (SLFHBA) is developed to automatically optimize model hyperparameters and initial weights for different datasets, reducing sensitivity to randomness and enabling stable, high-accuracy forecasting for individual wind and PV installations. The main innovations of this study are as follows:

- (1) A novel enhanced Honey Badger Algorithm, termed SLFHBA, is proposed by incorporating good point set principle, adaptive dynamic learning, and Gaussian mutation to enhance self-learning and adaptability.
- (2) A high-precision forecasting architecture is constructed by deeply coupling SLFHBA with Bidirectional Gated Recurrent Units to model renewable generation uncertainty and complex environmental interactions.
- (3) A multi-level robustness evaluation across seasons and weather conditions is established to verify predictive stability.
- (4) Multi-scenario validation across different energy types and sites demonstrates both high accuracy and strong generalization.
- (5) A new technical pathway for reliability evaluation of renewable energy forecasting models is provided.

The paper is organized as follows: Section 2 introduces the case and feature data; Section 3 describes the methodology; Section 4 develops the forecasting model; Section 5 conducts wind and PV case studies; and Section 6 concludes the paper.

Case and feature data

The case studies are based on the La Haute Borne wind farm in France and the DAKSC Alice Springs photovoltaic power plant in Australia. Fig. 1(a) presents the static information of La Haute Borne wind farm. The dataset includes wind power output, wind speed, wind direction, humidity, and other variables, with a temporal resolution of 10 min. To capture seasonal variability, data from three representative seasons, spring, autumn, and winter, were selected to assess model adaptability under different climatic conditions. According to wind turbine power curves and existing studies [33], wind speed, wind direction, and ambient temperature were chosen as input features, with wind power as the output. Data quality was ensured through outlier detection and interpolation, and all variables were standardized to remove dimensional effects, providing a reliable basis for subsequent modeling.

For PV power generation, this study adopts operational data from DAKSC Alice Springs photovoltaic power plant in Australia, as shown in Fig. 1(b). The dataset includes key parameters such as PV output power, ambient temperature, total horizontal irradiance, and relative humidity, with a temporal resolution of 5 min covering the effective generation period from 08:00 to 17:00 local time.

To objectively evaluate forecasting performance under different meteorological conditions, historical weather data were classified into three typical categories (sunny, cloudy, and rainy) using the K-means clustering method. Daily total irradiance, sunshine duration, cloud

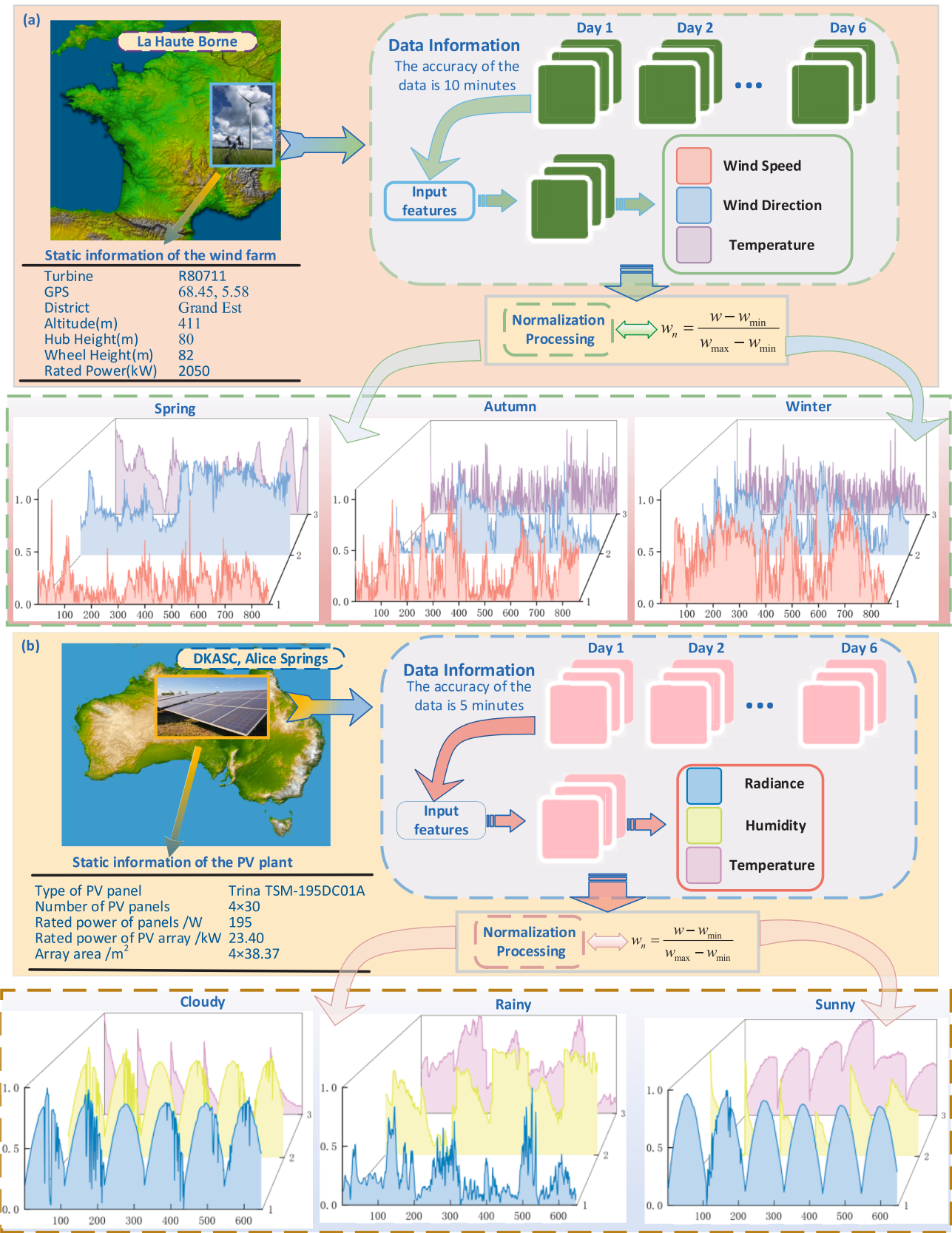


Fig. 1. Static information of wind farm and PV plant.

cover, and precipitation were selected as clustering features, and after standardization, the number of clusters was set to $K = 3$. The resulting clusters were interpreted according to their centroid characteristics to represent typical weather scenarios. Based on reference [34] and the physical characteristics of PV systems, irradiance, ambient temperature, and relative humidity were selected as input features, with PV output power as the output. This feature selection is consistent with the physical mechanism of PV generation and effectively captures the nonlinear relationship between meteorological conditions and power output. Data preprocessing followed the same procedure as for wind power, including outlier detection, interpolation, and variable standardization, ensuring data quality and model reliability.

Method

Bidirectional gated recurrent unit

The gated recurrent unit (GRU) is an improved recurrent neural network (RNN) architecture that employs update and reset gates to mitigate gradient vanishing and exploding problems. Compared with LSTM, GRU adopts a simpler structure with lower computational complexity while retaining strong temporal dependency modeling capability, making it well suited for short-term renewable energy forecasting.

To enhance multidimensional feature learning and capture spatio-temporal correlations in wind and solar sequences, this study extends the unidirectional GRU to a bidirectional GRU (BiGRU). By simultaneously learning forward and backward temporal dependencies, BiGRU provides a more comprehensive characterization of the structural patterns and dynamic evolution of time-series data across multiple time scales.

In GRU, the update and reset gates control information flow, updating the hidden state by combining historical states with current inputs via a candidate hidden state. This gating mechanism preserves key temporal features while remaining responsive to new information, thereby improving short-term forecasting performance. The BiGRU architecture further enhances temporal feature extraction by integrating forward and backward hidden states, enabling bidirectional dependency learning and more effective capture of long-term temporal patterns.

The forward hidden state $\vec{h}_t$ of BiGRU is represented as $GRU(x_t, \vec{h}_{t-1})$, and reverse hidden state $\overleftarrow{h}_t$ of BiGRU is represented as $GRU(x_t, \overleftarrow{h}_{t-1})$, where, x_t is the t -th element in the input sequence. And the final hidden state h_t can be represented as:

$$h_t = f(W_{\vec{h}_t} \vec{h}_t + W_{\overleftarrow{h}_t} \overleftarrow{h}_t + b_t) \quad (1)$$

where, $W_{\vec{h}_t}$ and $W_{\overleftarrow{h}_t}$ are the weight matrices corresponding to the forward and backward GRU layer states at time t , respectively; b_t is the bias of the hidden layer state at time.

Although BiGRU substantially enhances modeling capability, its predictive performance remains sensitive to key hyperparameter settings. Therefore, this study introduces a self-learning factor-based honey badger algorithm to optimize the network configuration and further improve forecasting accuracy.

Self-learning factors-based honey badger algorithm

The honey badger algorithm (HBA) is a nature-inspired optimization method that mimics the foraging behavior of honey badgers, incorporating digging and honey modes. To overcome the limitations of slow convergence and susceptibility to local optima in the original HBA, this study proposes a self-learning factors-based honey badger algorithm (SLFHBA), as described below.

To avoid uneven population distribution, which may affect the global search capability, the good point set is introduced into population

initialization to improve the distribution uniformity of honey badger populations within the feasible region.

In details, in the D -dimensional search space, generate a set of optimal points $Q_n(i) = \{[(v_1 \times i), (v_2 \times i), \dots, (v_l \times i), \dots, (v_D \times i)], 1 \leq i \leq n, 1 \leq l \leq D\}$ containing n points, where, $v_l = \{\cos(\frac{2\pi \times l}{l})\}$; $(v_l \times i)$ is the decimal part of $v_l \times i$; l is the smallest prime number that satisfies $2 \times D + 3 \leq l$.

The attraction intensity I_i is expressed as:

$$\begin{cases} I_i = r_2 \times \frac{S}{4\pi d_i^2} \\ S = (x_i - x_{i+1})^2 \\ d_i = x_{prey} - x_i \end{cases} \quad (2)$$

where, S is the resource density of honeycomb; d_i is the distance between the honeycomb and the i -th honey badger; x_{prey} is the honeycomb position.

More importantly, the density factor α dynamically decays with the number of iterations, effectively ensuring a smooth transition of SLFHBA from global exploration to local development. The foraging behavior of the honey badger population is regulated by the decision coefficient c , which takes a random value within the range of $[0,1]$ and has strong randomness characteristics. To optimize the transition process of the group between two search methods and achieve a smoother and more natural transition, an adaptive dynamic learning mechanism $c(t)$ is introduced in this study, its mathematically expressed as $(-\frac{t}{T_{max}} + 1) \times |(2 \times r_q - 1)|$, and r_q represents random number with a range of $[0, 1]$.

Finally, the position update of the honey badger population adopts a dual-mode mechanism, which is specifically regulated by the decision coefficient c . When $c < 0.5$, the population updates the position of the mining mode; when $c \geq 0.5$, switch to honey mode. The specific location update strategies in both modes are shown in equation (3).

$$x_{new} = \begin{cases} x_{prey} + F \times \beta \times I \times x_{prey} + F \times r_3 \times \alpha \times d_i \times [\cos(2\pi r_4) \times [1 - \cos(2\pi r_5)]] & c < 0.5 \\ x_{prey} + F \times r_6 \times \alpha \times d_i & c \geq 0.5 \end{cases} \quad (3)$$

where, x_{new} is the new location of honey badger; x_{prey} is the honeycomb position, which is the globally optimal solution found so far; β is a constant, indicating the ability of honey badgers to obtain honey; I is the attraction strength. W is used to change the search direction and expand the search range, if $r_7 \leq 0.5$, $W = 1$, On the contrary, $W = -1$.

After the position update is completed, to increase the population diversity in the algorithm optimization process and avoid the algorithm falling into local optima, Gaussian mutation strategy is introduced in the position update equation of the population.

The perturbation factor that follows a Gaussian distribution can be generated according to equation (4).

$$\begin{cases} y = \sqrt{2\sigma^2 \times \log(\frac{1}{1-u})} \times \cos(2\pi v) \\ u = rand \times (1 - (\frac{t}{T_{max}})^2) \end{cases} \quad (4)$$

where, v is a random number between 0 and 1; $\sigma = 0.1$; u is a random variable related to the number of iterations, ranging from 0 to 1.

The position update equation after introducing Gaussian mutation strategy is:

$$x_{new} = \begin{cases} \{x_{prey} + W \times \beta \times I \times x_{prey} + W \times r_3 \times \alpha \times d_i \times [\cos(2\pi r_4) \times [1 - \cos(2\pi r_5)]]\} y, c(l) < 0.5 \\ (x_{prey} + W \times r_6 \times \alpha \times d_i) y, c(l) \geq 0.5 \end{cases} \quad (5)$$

The iterative strategy and pseudo-code of SLFHBA are shown in Fig. 2.

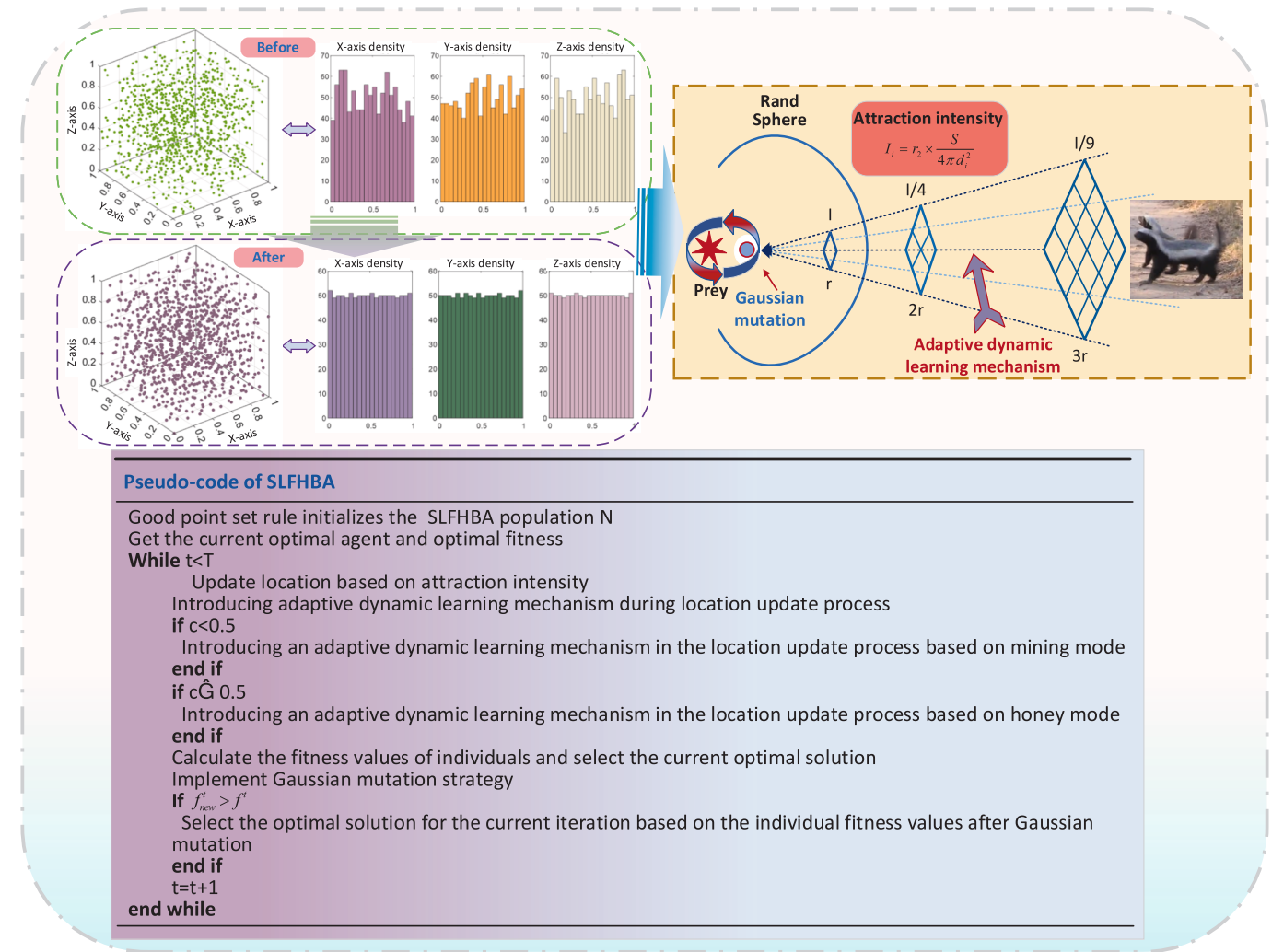


Fig. 2. Iterative strategy and pseudo-code of SLFHBA.

Descriptive statistical analysis

Benchmark functions serve as standardized tools for evaluating intelligent optimization algorithms. Following the testing framework in reference [35], this study constructs a representative benchmark function set with diverse modal characteristics to assess the global exploration and local exploitation abilities of the proposed algorithm. Detailed information on the selected functions, including dimensions, search ranges, and complexity, is provided in Table 1.

To ensure objective and comprehensive performance evaluation, a

systematic testing framework with multiple performance indicators is adopted. The proposed SLFHBA is compared with five representative metaheuristic algorithms, including particle swarm optimization (PSO), dwarf mongoose optimization algorithm (DMOA) [36], sparrow search algorithm (SSA) [37], tunicate swarm algorithm (TSA) [38], and the original HBA [39].

All algorithms are evaluated under identical conditions, including a population size of 30, a maximum of 500 iterations, and the same testing platform (Intel® Core™ i7-8655U CPU, 8 GB RAM, Matlab R2022a on Windows 10 64-bit), to ensure fair comparison. Each algorithm is

Table 1
Benchmark test function.

Attribute	Function	Range	Dimension	Optimal
unimodal	$f_1(u) = \sum_{i=1}^D u_i^2$	[-100, 100]	30	0
	$f_2(u) = \sum_{i=1}^D u_i + \prod_{i=1}^D u_i $	[-10, 10]	30	0
	$f_3(u) = \sum_{i=1}^D (\sum_{j=1}^i u_j)^2$	[-100, 100]	30	0
multimodal	$f_4(u) = \sum_{i=1}^D [u_i^2 - 10\cos(2\pi u_i) + 10D]$	[-5.12, 5.12]	30	0
	$f_5(u) = -20\exp(-0.2\sqrt{\frac{1}{D}\sum_{i=1}^D u_i^2}) - \exp(\frac{1}{D}\sum_{i=1}^D \cos(2\pi u_i)) + 20 + e$	[-32, 32]	30	0
	$f_6(u) = \frac{1}{4000}\sum_{i=1}^D u_i^2 - \prod_{i=1}^D \cos(\frac{u_i}{\sqrt{i}}) + 1$	[-600, 600]	30	0

independently executed 50 times, and the mean and standard deviation are reported in Table 2 to guarantee statistical reliability.

As shown in Table 2, SLFHBA demonstrates clear advantages on unimodal benchmark functions f_1 – f_3 . It converges to the theoretical global optimum of zero on f_1 and f_3 and achieves $9.47\text{E} - 215$ on f_2 . Moreover, SLFHBA yields the smallest standard deviations across all unimodal tests, indicating superior robustness and stability. These results confirm that SLFHBA achieves higher convergence accuracy and more reliable performance than the comparative algorithms on unimodal optimization problems.

In the optimization performance evaluation of multimodal test functions f_4 – f_6 the SLFHBA demonstrated excellent global optimization capability. In f_4 and f_6 , SLFHBA achieved optimal levels in convergence accuracy and algorithm stability. For f_5 , SLFHBA can still converge to a high-precision solution of $4.44\text{E}-16$.

The aforementioned statistical analysis results fully demonstrate that the SLFHBA exhibits significant performance advantages in solving optimization problems. Experimental data confirm that SLFHBA possesses outstanding global search capability, effectively avoiding local optima while demonstrating excellent optimization robustness.

Establishment of a short-term forecasting model based on hybrid artificial intelligence and deep learning

This study combines intelligent optimization algorithms with deep learning to develop a high-precision wind and PV power forecasting framework. The proposed SLFHBA is used to optimize the hyper-parameters of a BiGRU-based model, aiming to minimize forecasting errors and accurately capture power output characteristics. By adaptively tuning model parameters, SLFHBA significantly improves convergence speed and generalization performance. The hybrid forecasting framework and its implementation process for wind and PV case studies are illustrated in Fig. 3, with the main steps summarized as follows:

Step 1: obtain historical power data and corresponding meteorological data of wind and PV power, conduct a preliminary evaluation of the integrity of the data and the continuity of the time series, and

ensure that the dataset can meet the needs of model training and validation.

Step 2: identify abnormal data points based on physical thresholds, use interpolation or deletion strategies for correction, and standardize the corrected data to provide a high-quality data foundation for model training.

Step 3: set the internal parameters and determine the structure of the model, and initialize the model parameters.

Step 4: use the mean squared error (MSE) between the actual and predicted values of wind and PV power generation as the fitness function for the SLFHBA algorithm, and start iterative optimization.

Step 5: train the SLFHBA-BiGRU model and use SLFHBA iterative optimization to obtain the optimal parameters configuration.

Step 6: based on the optimal parameters combination, use the model to predict the test set and obtain the predicted results of wind and PV power generation.

Step 7: evaluate the performance of the SLFHBA-BiGRU model in wind and PV power test sets based on root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) and goodness of fit (R2). where, Q represents the number of test sample points; $f(x_i)$ is the forecasting value; Y_i is the actual value.

Case-based analysis

Case 1 – forecasting results analysis of wind power

Based on the actual operating data of the wind farm described in Section 2, this study conducts empirical analyses of wind power forecasting across multiple seasons. To comprehensively evaluate model performance, six consecutive days of operational data (864 samples per season) from three representative seasons of spring, autumn, and winter were selected. The first five days of data (720 samples) were used for model training, while the sixth day (144 samples) served as an independent test set. For performance comparison, the proposed SLFHBA-BiGRU was benchmarked against the GRU, PSO-BiGRU, and HBA-BiGRU models.

Fig. 4 compares the forecasting performance of the four models in spring, autumn, and winter, including forecasting curves and fitting results. As shown in Fig. 4(a), the GRU model exhibits notable deviations between predicted and actual values, mainly due to the limited temporal feature extraction capability of its unidirectional architecture, which constrains forecasting accuracy.

Under spring conditions, PSO-BiGRU and HBA-BiGRU show improved forecasting accuracy, particularly in the initial prediction stage. The proposed SLFHBA-BiGRU achieves the best overall performance and accurately tracks wind power variations even during high-fluctuation periods (11:00–13:00). In contrast, the GRU model exhibits large deviations from actual values in both autumn and winter forecasting tasks.

As shown in Fig. 4(b), SLFHBA-BiGRU closely fits the actual wind power output, with prediction points densely clustered around the fitting curve and symmetrically distributed on both sides, indicating strong fitting capability. Overall, Fig. 4 confirms the superior forecasting accuracy and robustness of SLFHBA-BiGRU, demonstrating the effectiveness of the proposed SLFHBA algorithm in parameter optimization.

Case 2 – forecasting results analysis of PV power

This study performs empirical PV power forecasting under multiple weather conditions using the actual operational data described in Section 2. A weather stratified sampling strategy was adopted, selecting six consecutive days of data under three typical conditions (sunny, cloudy, and rainy). The first five days were used for training, and the sixth day for testing. The predictive performance of the proposed SLFHBA-BiGRU was compared with GRU, PSO-BiGRU, and HBA-BiGRU, with results

Table 2
Statistical results of algorithm optimization.

f	PSO		DMOA		SSA	
	Avg	Std	Avg	Std	Avg	Std
f_1	1.33E-20	5.62E-20	5.98	2.58	3.54E-07	1.21E-06
f_2	2.00E-11	3.23E-11	13.79	17.21	2.27	1.86275
f_3	3.19E-06	5.70E-06	6.31E+04	1.19E+04	1.53E+03	8.85E+02
f_4	4.78	2.39	2.33E+02	17.90	51.72	18.22
f_5	4.49E-11	6.08E-11	3.13	0.57	2.38	0.78
f_6	0.22	0.18	1.05	0.02	0.02	0.01
f	TSA		HBA		SLFHBA	
	Avg	Std	Avg	Std	Avg	Std
f_1	2.74E-21	8.77E-21	3.04E-128	1.99E-127	0	0
f_2	7.89E-14	9.16E-14	1.32E-68	4.98E-68	9.47E-215	0
f_3	7.96E-4	0.002085	3.00E-92	2.08E-91	0	0
f_4	1.87E+02	38.35	0	0	0	0
f_5	1.77	1.56	1.59	5.46	4.44E-16	0
f_6	9.8E-03	1.51-02	0	0	0	0

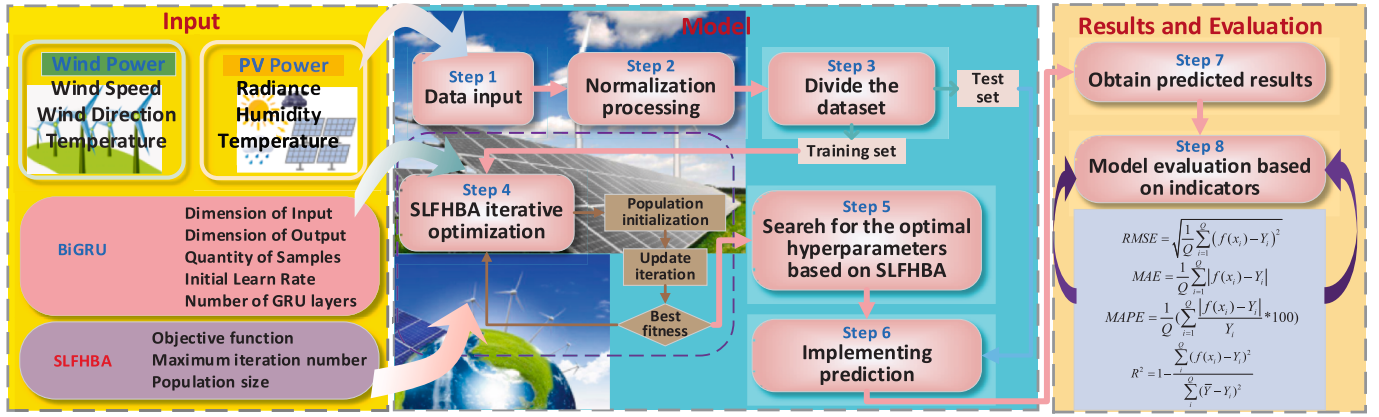


Fig. 3. The process of hybrid artificial intelligence and deep learning forecasting model.

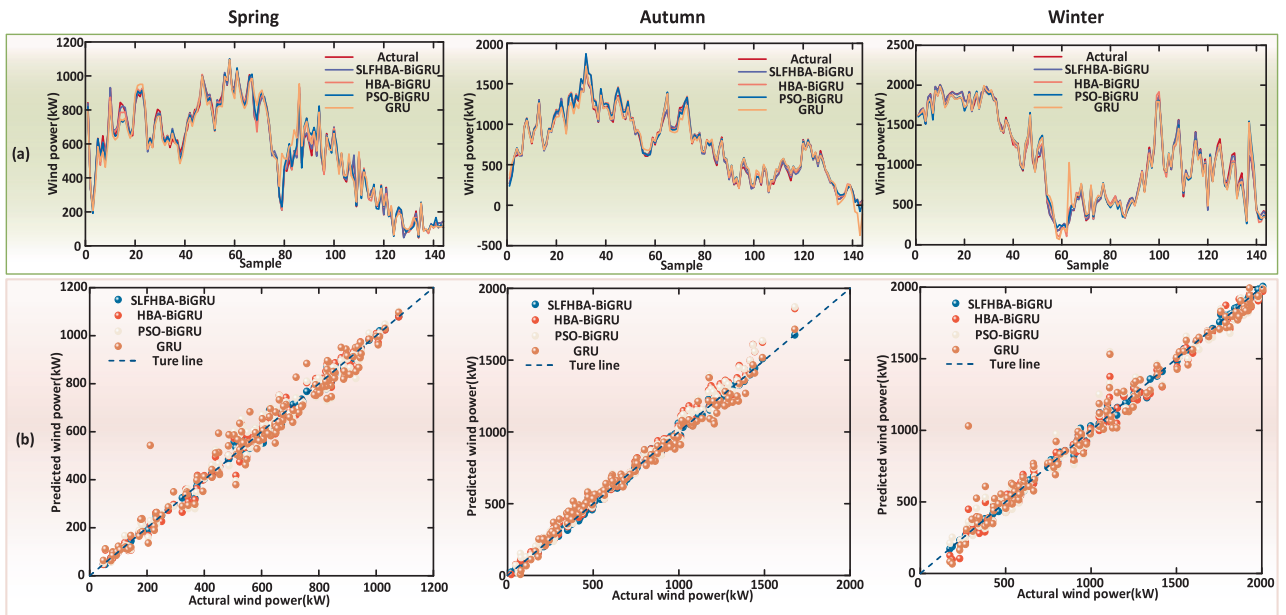


Fig. 4. The forecasting results of wind power for case 1.

presented in Fig. 5.

Under sunny weather condition, Fig. 5(a) shows that GRU, PSO-BiGRU, HBA-BiGRU and SLFHBA-BiGRU model all demonstrate good fitting performance with the actual power curve, indicating their strong predictive capabilities for time-series data. The scatter plot analysis in Fig. 5(b) shows that the predicted power value of the SLFHBA-BiGRU model is highly consistent with the actual value, and its scatter points are closely distributed around the fitted curve. Under cloudy and rainy conditions, all models perform well in the initial forecasting stage; however, GRU exhibits growing errors over time, and PSO-BiGRU and HBA-BiGRU show noticeable error fluctuations. In contrast, SLFHBA-BiGRU maintains consistent agreement with actual outputs throughout the forecasting horizon, with prediction points tightly and uniformly distributed around the fitted curve, confirming its superior accuracy and robustness under complex weather conditions.

Statistical results of performance indicators

Based on evaluation metrics, the wind and PV power forecasting performance of each model across different seasons was quantitatively assessed, as summarized in Table 3. In spring, autumn, and winter, SLFHBA-BiGRU consistently outperforms the comparison models,

achieving the lowest forecasting errors and highest fitting accuracy. Compared with HBA-BiGRU, SLFHBA-BiGRU reduces RMSE by 13.03 kW, 31.82 kW, and 34.27 kW, decreases MAPE by 11.89 %, 4.64 %, and 4.66 %, and increases R2 by 0.84 %, 0.90 %, and 0.90 %, respectively. Although SLFHBA-BiGRU requires longer runtime due to its more refined global search strategy for hyperparameter optimization, this trade-off leads to significantly improved prediction accuracy and demonstrates strong stability and cross-season robustness.

Under sunny conditions, GRU achieves a MAPE of 0.89 % and an R2 of 99.85 %, while PSO-BiGRU and HBA-BiGRU further improve performance, reducing MAPE to 0.52 % and 0.48 % and increasing R2 to 99.94 % and 99.95 %, respectively. Compared with HBA-BiGRU, SLFHBA-BiGRU further reduces MAPE by 0.31 % and improves R2 by 0.04 %, indicating superior overall forecasting performance. Although SLFHBA-BiGRU incurs higher computational cost, this is consistent with the wind power case and is considered acceptable in short-term renewable energy forecasting, where accuracy is the primary objective.

Through systematic testing across multiple weather conditions, this study has validated the generalizability and robustness characteristics of the SLFHBA-BiGRU model. Quantitative analysis results demonstrate that, compared with other models, the SLFHBA-BiGRU model exhibits significant predictive performance advantages, exceptional cross-

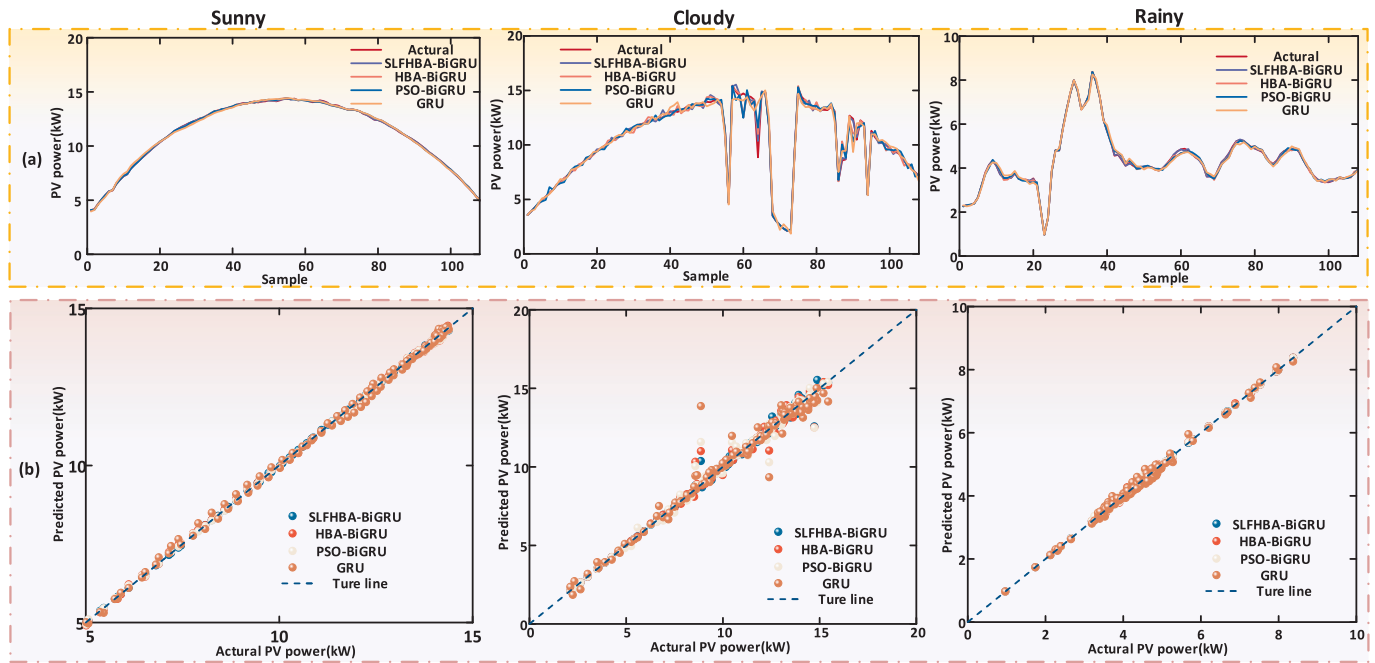


Fig. 5. The forecasting results of PV power for case 2.

Table 3

Statistical results of performance indicators for different models.

Type	Season	Model	Index				
			RMSE (kW)	MAE (kW)	MAPE (%)	R2 (%)	Time (s)
Wind power	Spring	GRU	54.32	37.81	9.49	96.02	60.58
		PSO-BiGRU	37.79	27.82	6.97	98.07	150.83
		HBA-BiGRU	30.49	23.15	5.65	98.74	160.12
		SLFHBA-BiGRU	17.46	11.26	2.11	99.58	311.83
	Autumn	GRU	67.91	48.23	26.66	96.99	66.06
		PSO-BiGRU	56.13	42.18	11.86	98.49	151.20
		HBA-BiGRU	54.26	39.47	7.62	98.77	161.09
		SLFHBA-BiGRU	22.44	17.19	2.98	99.67	307.45
	Winter	GRU	96.39	58.89	9.35	96.92	61.11
		PSO-BiGRU	74.99	55.30	6.98	98.15	148.72
		HBA-BiGRU	59.07	46.90	6.57	98.89	157.43
		SLFHBA-BiGRU	24.80	15.91	1.91	99.79	306.75
PV power	Sunny	GRU	0.11	0.09	0.89	99.85	55.23
		PSO-BiGRU	0.07	0.05	0.52	99.94	115.29
		HBA-BiGRU	0.06	0.05	0.48	99.95	123.65
		SLFHBA-BiGRU	0.03	0.02	0.17	99.99	302.69
	Cloudy	GRU	0.68	0.34	3.77	96.20	58.04
		PSO-BiGRU	0.49	0.25	2.41	98.01	112.52
		HBA-BiGRU	0.43	0.23	2.03	98.51	119.98
		SLFHBA-BiGRU	0.32	0.16	1.36	99.18	311.97
	Rainy	GRU	0.11	0.09	2.03	99.27	56.99
		PSO-BiGRU	0.08	0.06	1.39	99.61	119.81
		HBA-BiGRU	0.07	0.05	1.24	99.67	127.14
		SLFHBA-BiGRU	0.02	0.01	0.31	99.97	301.76

scenario adaptability and reliable predictive robustness.

Concluding remarks

Existing short-term forecasting methods predominantly focus on individual energy sources and still require improvements in accuracy when handling variable weather conditions. To address this critical challenge, a novel forecasting model is proposed in this study, the main conclusions are as follows:

- (1) The proposed SLFHBA-BiGRU model has demonstrated excellent performance in the task of forecasting renewable energy

generation. Compared with the benchmark GRU model, in the wind power prediction task, the RMSE of BiGRU optimized based on SLFHBA decreased by 36.86 kW, 45.47 kW and 71.59 kW respectively in three typical seasons, and the MAPE decreased by 7.38 %, 23.68 % and 7.44 %; in the photovoltaic power prediction task, under three typical weather conditions, the RMSE of BiGRU optimized based on SLFHBA decreased by 0.08 kW, 0.36 kW and 0.09 kW, and the MAPE decreased by 0.72 %, 2.41 % and 1.72 %. Under different seasons and weather conditions, the prediction error of the proposed model remains at a low level, demonstrating good environmental adaptability and generalization ability.

- (2) Compared with existing algorithms, the SLFHBA algorithm has improved convergence accuracy and stability in unimodal and multimodal test function optimization problems, demonstrating outstanding advantages. Specifically, in f_1 , f_3 , f_4 and f_6 , it can converge to the theoretical optimal value of 0 and exhibit stronger robustness. In addition, the BiGRU model optimized based on SLFHBA has shown significant advantages in evaluation indexes in renewable energy forecasting tasks. Compared to HBA-BiGRU, in the task of wind power prediction, compared to HBA-BiGRU, the RMSE of the SLFHBA-BiGRU model decreased by 13.03 kW, 31.82 kW and 34.27 kW respectively, MAPE decreased by 11.89 %, 4.64 % and 4.66 %; in the task of PV power prediction, the RMSE of the SLFHBA-BiGRU model decreased by 0.03 kW, 0.11 kW and 0.05 kW respectively, MAE decreased by 0.03 kW, 0.07 kW and 0.045 kW.
- (3) This study designed a systematic validation framework that covers the characteristics of spring, autumn and winter, and the meteorological conditions of sunny, cloudy and rainy. It constructed a testing system that can comprehensively evaluate the accuracy and generalization of forecasting model, achieving innovation in model validation methods. Through wind and PV power forecasting in different regions, this study has been confirmed that the model significantly improves forecasting accuracy and universality. It provides a reliable performance evaluation paradigm for the engineering application of renewable energy prediction models.

The contributions and innovations of this study can be summarized as follows: (1) a hybrid deep learning model that integrates SLFHBA and BiGRU is established, which can be used for wind and PV power short-term forecasting; (2) a new SLFHBA algorithm considering individual

self-learning and dynamic adjustment capabilities has been proposed, demonstrating excellent convergence performance and the ability to escape local optima; (3) innovatively designed a robust validation scheme across seasons and weather conditions, proving the predictive accuracy and universality of the proposed model from a statistical perspective.

Although the proposed model achieves excellent forecasting performance, the current validation is conducted on specific turbine types and geographic locations; therefore, the trained model is not intended to be directly transferable without retraining. Accordingly, future work will focus on validating the proposed methodology across diverse wind farms and PV plants, as well as exploring its performance under limited data conditions and in combination with transfer learning techniques.

CRedit authorship contribution statement

Jiaqi Liu: Writing – original draft, Methodology, Conceptualization. **Yuwei Liu:** Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Qiang Shen:** Writing – original draft, Methodology. **Keyan Li:** Writing – original draft. **Yinuo Chen:** Writing – original draft.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Nomenclature

BiGRU	Bidirectional gated recurrent unit
DMOA	Dwarf mongoose optimization algorithm
ELM	Extreme learning machine
GP	Gaussian process
GRU	Gated recurrent unit
HBA	Honey badger algorithm
LSTM	Long short term memory
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MSE	Mean squared error
NWP	Numerical weather forecasting
PSO	Particle swarm optimization
PV	Photovoltaic
RNN	Recurrent neural network
RMSE	Root mean squared error
SLFHBA	Self-learning factors-based honey badger algorithm
SSA	Sparrow search algorithm
SVM	Support vector machine
TSA	Tunicate swarm algorithm

Data availability

Data will be made available on request.

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